

Effects of expanding higher education vacancies: effects on marginal and infra-marginal students

Preliminary Results

28 de septiembre de 2026

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Data and sample

- We combine DEMRE and MINEDUC administrative records for admission cohorts 2007–2016.
- The resulting database links applications, enrollment outcomes, graduation outcomes, and PSU scores at the applicant level.
- The main analysis focuses on applicants within 25 points of the regular admission cutoff.

Statistic	Sample	N	Mean
Applications	Full sample	4,434,346	–
Unique students	Full sample	786,071	–
Program-years	Full sample	11,789	–
Running variable	BW25	839,108	-2.326
Above cutoff	BW25	839,108	0.433
Enrolls target	BW25	839,108	0.369
Enrolls university	BW25	839,108	0.833
Enrolls HE	BW25	839,108	0.870
Graduates target (8y)	Grad. sample	838,307	0.140
Graduates university (8y)	Grad. sample	838,307	0.402
Graduates HE (8y)	Grad. sample	838,307	0.423

RDD around admission cutoffs

- Running variable: application score centered around the regular admission cutoff,

$$r_{ipt} = score_{ipt} - cutoff_{pt}.$$

- Treatment indicator:

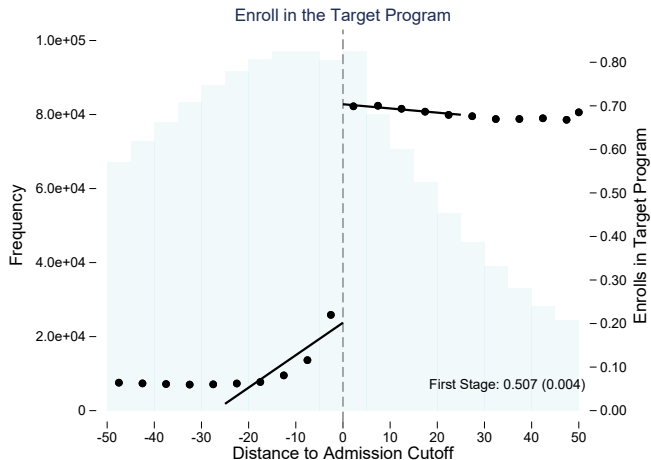
$$A_{ipt} = 1(r_{ipt} \geq 0).$$

- We compare applicants marginally above and below the cutoff within the same program-year.
- Main specification:

$$Y_{ipt} = \beta A_{ipt} + f(r_{ipt}) + \mu_{pt} + \varepsilon_{ipt}$$

Main bandwidth: $|r_{ipt}| \leq 25$. Standard errors are clustered at the program-year level.

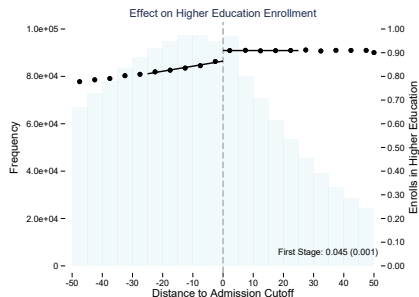
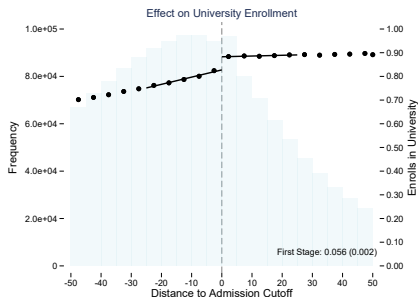
Enrollment in the Target Program



RDD estimate
0.507 (0.004)

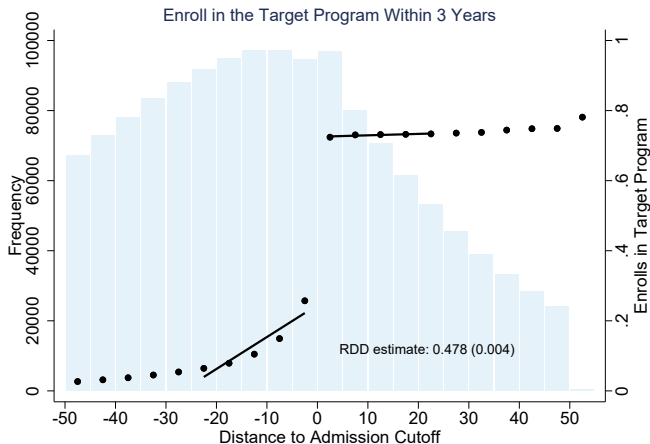
Crossing the cutoff
increases
target-program
enrollment by
approximately 51
percentage points.

Enrollment on University and HE Program



- Most applicants below the cutoff still access university or higher education through alternative options.
- Crossing the cutoff mainly changes **where students enroll**, rather than **whether they enroll**.

Target Enrollment Within Three Years

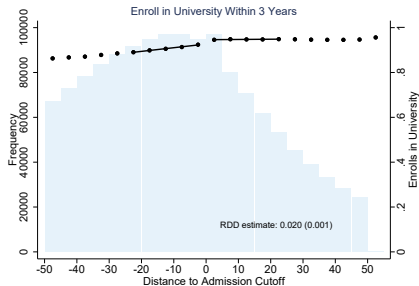


Regression BW = ± 25 . Plot range = ± 50 . FE: program-year. SE clustered by program-year. N=839108, clusters=8369.

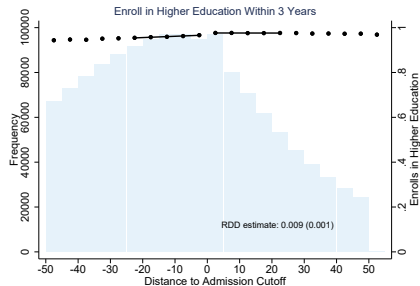
RDD estimate
0.478
(0.004)

Crossing the cutoff increases target-program enrollment within three years by approximately 48 percentage points.

Broader Enrollment Outcomes Within Three Years



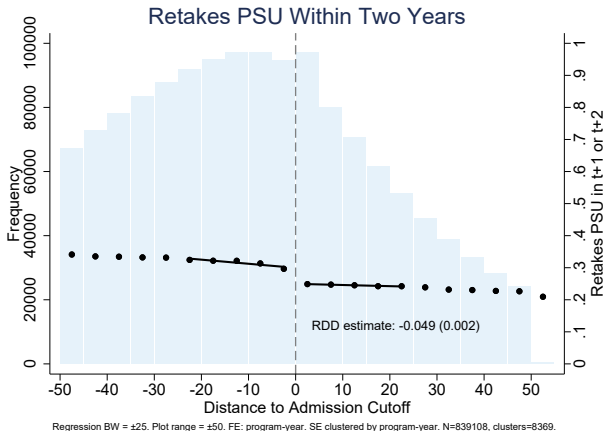
Regression BW = ± 25 . Plot range = ± 50 . FE: program-year. SE clustered by program-year. N=839108, clusters=8369.



Regression BW = ± 25 . Plot range = ± 50 . FE: program-year. SE clustered by program-year. N=839108, clusters=8369.

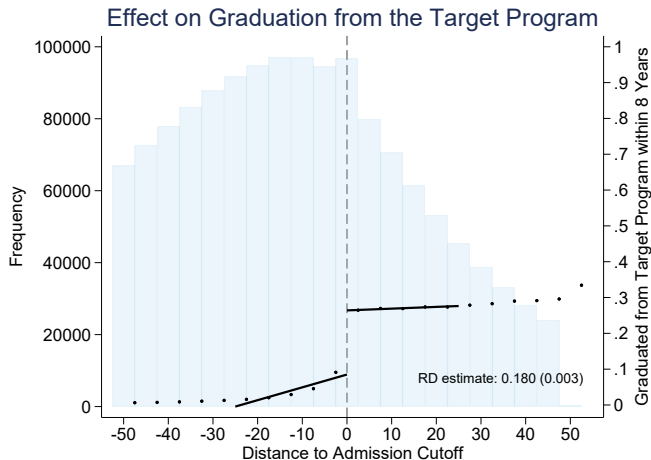
- Most applicants below the cutoff still access university or higher education through alternative options.
- The effect on overall higher-education participation remains small.
- The main effect continues to operate through the specific program attended.

PSU Retaking Within Two Years



- Crossing the cutoff reduces PSU retaking within the next two years.
- Applicants above the cutoff are less likely to re-enter the admission process.
- Admission eligibility lowers incentives to improve future admission outcomes.

Target Graduation Within Eight Years

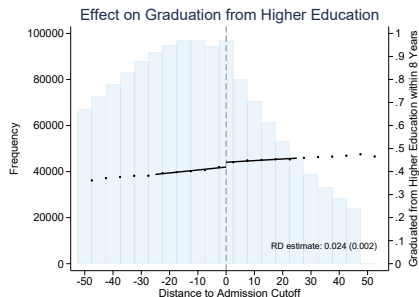
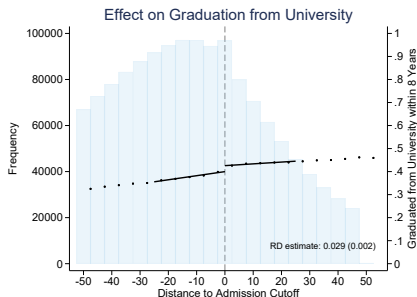


RDD estimate
0.180
(0.003)

Crossing the cutoff
increases graduation
from the target
program by
approximately 18
percentage points.

Graduation is measured within eight years of the application year.

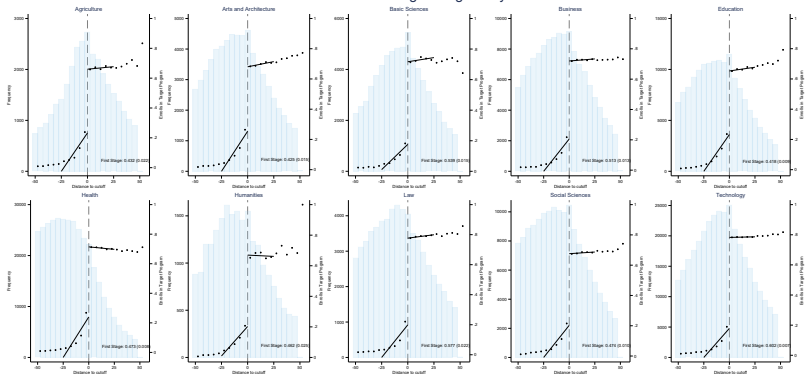
Broader Graduation Outcomes Within Eight Years



- Broader graduation effects are positive but much smaller than the target-program effect.
- Crossing the cutoff mainly changes where students graduate, rather than whether they complete higher education.

Target Enrollment Effects by Field

RDD: Enroll in the Target Program by Field



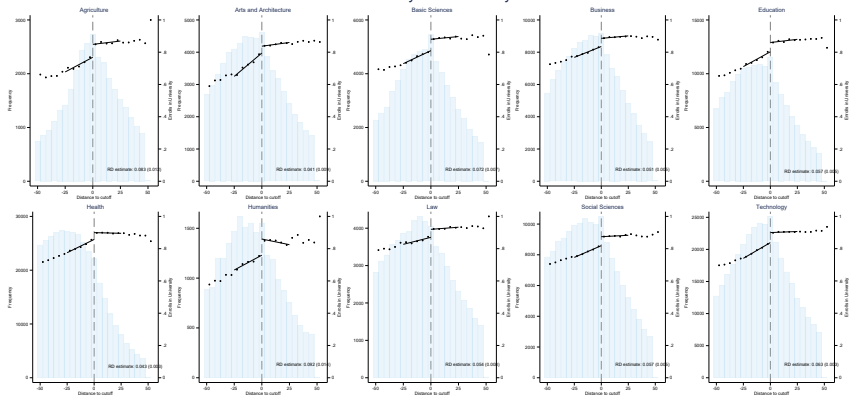
- Crossing the cutoff increases target-program enrollment in every field.
- Effects are largest in Technology, Law, Basic Sciences, and Business.

Target Enrollment Effects by Field

Field	Target	University	HE	N
Agriculture	0.432	0.083	0.065	21,102
Arts & Architecture	0.425	0.041	0.034	39,715
Basic Sciences	0.539	0.072	0.063	43,849
Business	0.513	0.051	0.034	79,964
Education	0.418	0.057	0.044	95,508
Health	0.473	0.043	0.039	206,050
Humanities	0.462	0.092	0.084	13,470
Law	0.577	0.054	0.054	37,000
Social Sciences	0.474	0.057	0.049	90,504
Technology	0.602	0.063	0.047	211,145

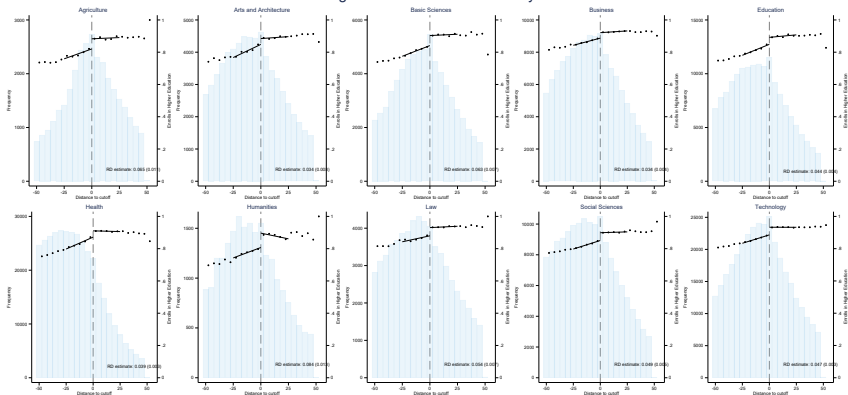
University enrollment effects by field.

RDD: University Enrollment by Field



Higher-education enrollment effects by field

RDD: Higher Education Enrollment by Field

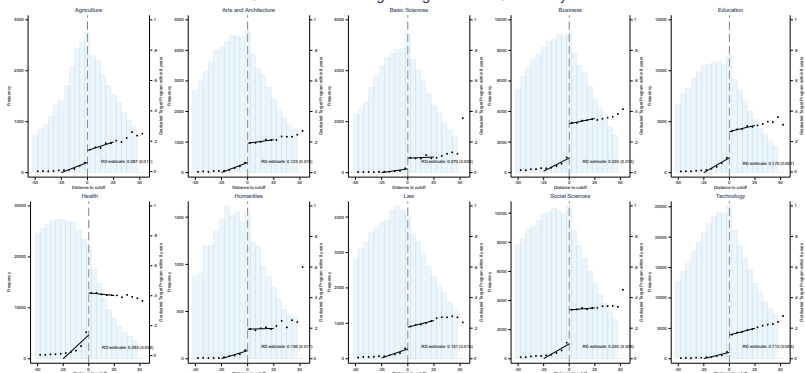


RDD estimates by field: university and HE enrollment

Field	Outcome	Estimate	SE	N	Baseline
Agriculture	University	0.083	0.012	21,102	0.729
	HE	0.065	0.011	21,102	0.793
Arts & Architecture	University	0.041	0.009	39,715	0.723
	HE	0.034	0.008	39,715	0.808
Basic Sciences	University	0.072	0.007	43,849	0.771
	HE	0.063	0.007	43,849	0.809
Business	University	0.051	0.005	79,964	0.802
	HE	0.034	0.004	79,964	0.865
Education	University	0.057	0.005	95,508	0.757
	HE	0.044	0.004	95,508	0.815
Health	University	0.043	0.003	206,050	0.815
	HE	0.039	0.003	206,050	0.834
Humanities	University	0.092	0.014	13,470	0.713
	HE	0.084	0.013	13,470	0.775
Law	University	0.054	0.008	37,000	0.849
	HE	0.054	0.007	37,000	0.862
Social Sciences	University	0.057	0.005	90,504	0.782
	HE	0.049	0.005	90,504	0.824
Technology	University	0.063	0.003	211,145	0.790
	HE	0.047	0.003	211,145	0.860

Target Graduation Effects by Field

RDD: Graduation from Target Program within 8 Years by Field



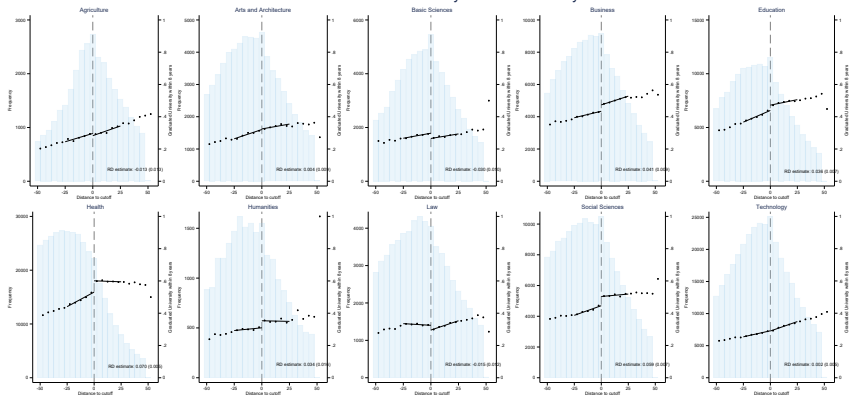
- Crossing the cutoff increases target-program graduation across all fields.
- The largest effects appear in Health, Business, Social Sciences, Education, and Law.

RDD estimates by field: target graduation

Field	Estimate	SE
Agriculture	0.087	0.011
Arts & Architecture	0.123	0.010
Basic Sciences	0.070	0.008
Business	0.224	0.013
Education	0.170	0.007
Health	0.283	0.008
Humanities	0.136	0.017
Law	0.151	0.016
Social Sciences	0.224	0.009
Technology	0.113	0.004

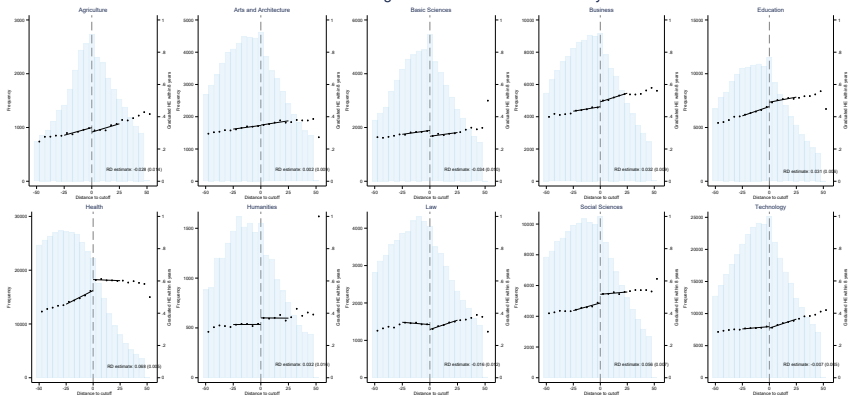
Graduation effects by field: university

RDD: Graduation from University within 8 Years by Field



Graduation effects by field: any higher education

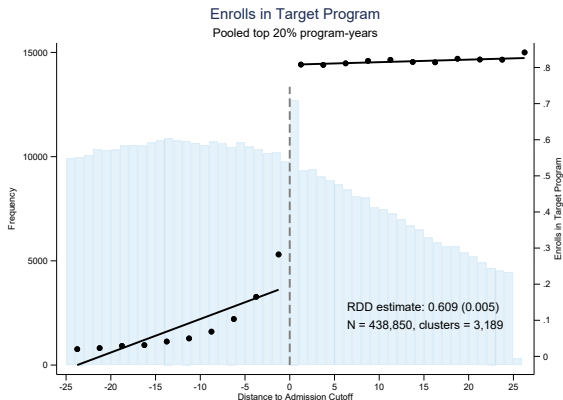
RDD: Graduation from Higher Education within 8 Years by Field



RDD estimates by field: university and HE graduation

Field	Outcome	Estimate	SE	N	Baseline
Agriculture	University	-0.013	0.013	21,102	0.274
	HE	-0.028	0.014	21,102	0.312
Arts & Architecture	University	0.004	0.009	39,715	0.292
	HE	0.002	0.009	39,715	0.335
Basic Sciences	University	-0.030	0.010	43,849	0.283
	HE	-0.034	0.010	43,849	0.303
Business	University	0.041	0.009	79,964	0.414
	HE	0.032	0.009	79,964	0.448
Education	University	0.036	0.007	95,508	0.403
	HE	0.031	0.006	95,508	0.434
Health	University	0.070	0.005	206,050	0.484
	HE	0.068	0.005	206,050	0.497
Humanities	University	0.034	0.016	13,470	0.301
	HE	0.032	0.016	13,470	0.330
Law	University	-0.015	0.012	37,000	0.330
	HE	-0.016	0.012	37,000	0.337
Social Sciences	University	0.059	0.007	90,504	0.417
	HE	0.056	0.007	90,504	0.440
Technology	University	0.002	0.005	211,145	0.273
	HE	-0.007	0.005	211,145	0.310

Selective Programs: Target Enrollment

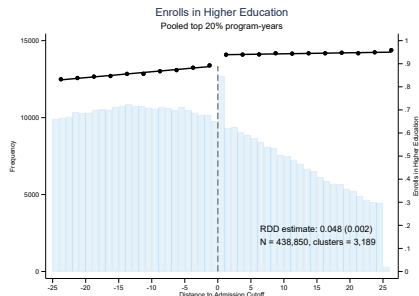
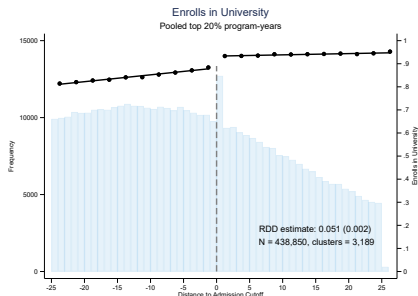


Selective programs

- Selectivity is based on the Language–Math score distribution of admitted students.
- Programs are classified as selective if they appear at least once in the top 20 % of the pooled post-2012 selectivity ranking.

Crossing the cutoff in selective programs increases target-program enrollment by approximately 61 percentage points.

Selective Programs: Broader Enrollment Outcomes



- Effects on university and higher-education enrollment remain much smaller than target-program enrollment.
- Even among selective programs, cutoff crossing mainly changes **where students enroll**, not overall higher-education participation.

Construction of Delta Selectivity

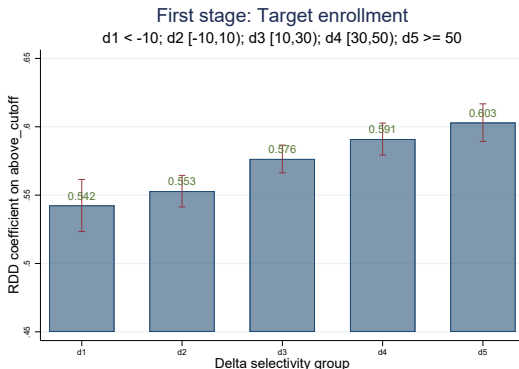
- For each target application, we identify the student's next-best feasible alternative.
- Program selectivity is measured by the average Language–Math score among regularly admitted students.
- We define:

$$\Delta Selectivity = Selectivity_{target} - Selectivity_{nextbest}.$$

- Positive values indicate that the target program is more selective than the next-best alternative.

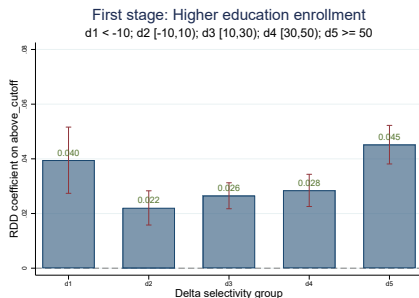
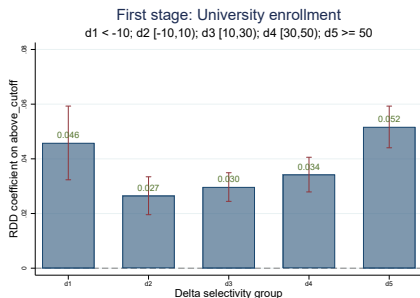
Group	Definition
d1	$\Delta < -10$
d2	$[-10, 10)$
d3	$[10, 30)$
d4	$[30, 50)$
d5	≥ 50

Target Enrollment by Selectivity Gap



- Crossing the cutoff increases target-program enrollment across all groups.
- Effects are strongest when the target program is substantially more selective than the next-best alternative.
- Admission matters most when students face large differences in program selectivity.

Broader Enrollment Outcomes by Selectivity Gap

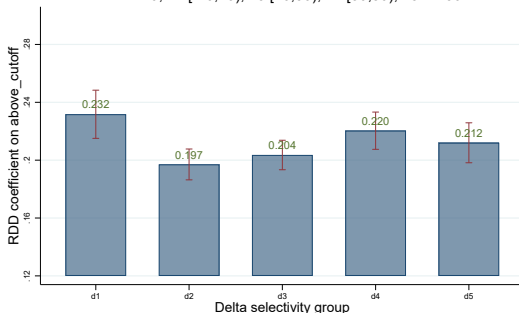


- Effects on university and higher-education enrollment remain relatively small across groups.
- Differences across selectivity-gap groups are much less pronounced than for target-program enrollment.
- The main response margin remains the specific program attended rather than overall higher-education participation.

Target Graduation by Selectivity Gap

Graduation from target program within 8 years

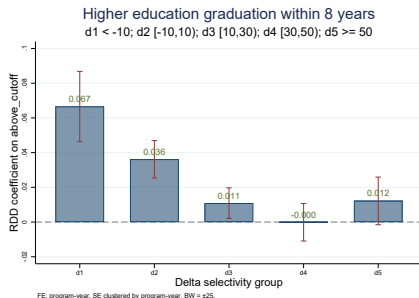
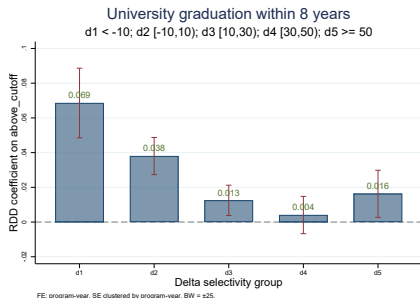
d1 < -10; d2 [-10,10); d3 [10,30); d4 [30,50); d5 >= 50



FE: program-year. SE clustered by program-year. BW = ± 25 .

- Crossing the cutoff increases target-program graduation across all selectivity-gap groups.
- Target-graduation effects remain large across the distribution of target versus next-best selectivity.

Broader Graduation Outcomes by Selectivity Gap

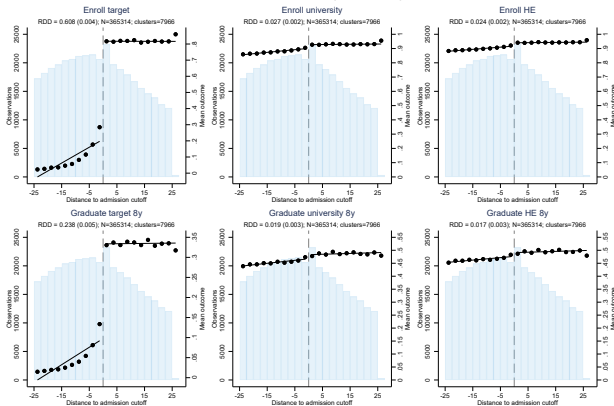


- Effects on university and higher-education graduation are smaller than target-program graduation effects.
- Broader graduation effects are more heterogeneous across selectivity-gap groups.
- Crossing the cutoff mainly changes where students graduate, rather than whether they complete higher education.

RDD around cutoff: same field

RDD around cutoff: Same field

Binned means, local linear fits, and running-variable histogram. BW = ± 25 .

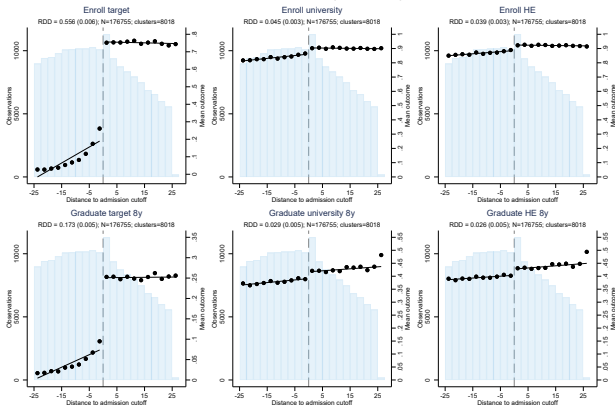


- Crossing the cutoff generates large increases in target-program enrollment and graduation when the next-best alternative belongs to the same field.
- Effects on broader university and higher-education outcomes remain small, suggesting that admission mainly affects program choice within a field.

RDD around cutoff: different field

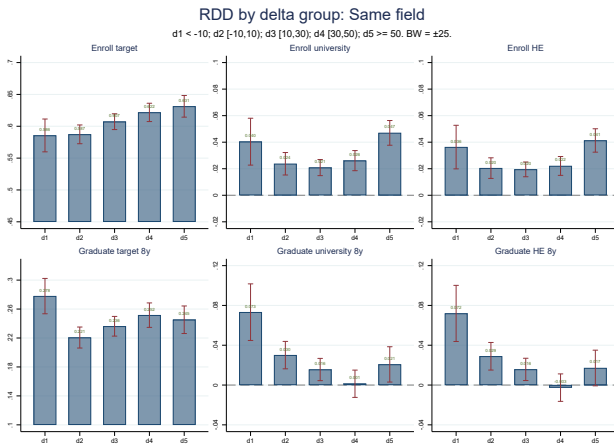
RDD around cutoff: Different field

Binned means, local linear fits, and running-variable histogram. BW = ± 25 .



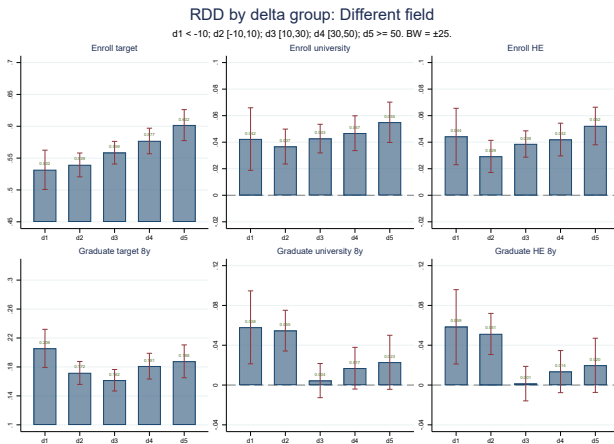
- Target-program enrollment and graduation effects remain large when the next-best alternative belongs to a different field.
- Effects on broader university and higher-education outcomes are somewhat larger than in the same-field sample, consistent with admission affecting both program and field choice.

RDD by selectivity gap: same field



- Within same-field alternatives, target-enrollment effects increase with the selectivity gap between the target and next-best program.
- Broader university and higher-education graduation effects remain comparatively small across groups.

RDD by selectivity gap: different field



- Within different-field alternatives, target-enrollment effects also increase with the selectivity gap.
- Broader graduation effects are less stable across groups, suggesting more heterogeneous substitution patterns when the fallback option is in another field.

From Marginal to Inframarginal Students

- The previous analysis focused on applicants located around admission cutoffs.
- We now study **inframarginal students**: applicants who would have been admitted to their target program in every year of the relevant period.
- To identify them, we reconstruct each student's application score using the admission weights from each year and compare it with the corresponding admission cutoff.
- The objective is to estimate whether changes in program capacity affect the outcomes of these inframarginal cohorts.

Specification

- Following the model notes, program size should be measured using *total enrollment*:

$$\Delta N_{pt}^{total} = \pi \Delta Z_{pt} + \alpha_t + \nu_{pt}.$$

- Outcomes continue to be computed using inframarginal students:

$$\Delta y_{pt}^{infra} = \beta \widehat{\Delta N}_{pt}^{total} + \alpha_t + \varepsilon_{pt}.$$

- N_{pt}^{total} : total enrollment in program p and year t .
- Z_{pt} : total vacancies/cupos.
- y_{pt}^{infra} : graduation rate among inframarginal students.

Corrected Inframarginal IV Results

Period	Instrument	Outcome	First stage	F-stat	Reduced form	2SLS
2007–2012	Δ Total Cupos	Target grad. 8y	0.816 (0.037)	482.57	-0.00038 (0.00024)	-0.00048 (0.00030)
2007–2012	Δ Total Cupos	Univ. grad. 8y	0.816 (0.037)	482.57	0.00063 (0.00030)	0.00079** (0.00035)
2007–2012	Δ Total Cupos	HE grad. 8y	0.816 (0.037)	482.57	0.00063 (0.00030)	0.00079** (0.00035)
2013–2016	Δ Total Cupos	Target grad. 8y	0.543 (0.054)	96.11	-0.00011 (0.00034)	-0.00020 (0.00064)
2013–2016	Δ Total Cupos	Univ. grad. 8y	0.543 (0.054)	96.11	0.00048* (0.00025)	0.00088** (0.00039)
2013–2016	Δ Total Cupos	HE grad. 8y	0.543 (0.054)	96.11	0.00048* (0.00025)	0.00088** (0.00039)

Standard errors in parentheses. First stage: ΔN_{pt}^{total} on ΔZ_{pt} . Reduced form: Δy_{pt}^{infra} on ΔZ_{pt} . 2SLS: Δy_{pt}^{infra} on predicted ΔN_{pt}^{total} . * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$.

First Stage Robustness and Residual Variation

Panel A. Sequential first-stage specifications

Specification	2007–2012	2013–2016
Differences, no FE	0.817	0.543
Differences + year FE	0.816	0.543
Differences + program FE	0.816	0.637
Differences + program + year FE	0.813	0.638
Levels + program + year FE	0.824	0.525

Panel B. Residual variation after program and year FE

Period	Enrollment SD	Residual enrollment SD	Cupos SD	Residual Cupos SD
2007–2012	45.5	31.8	49.6	33.5
2013–2016	62.4	37.9	67.7	40.0

- Adding fixed effects does not eliminate the first stage.
- Residual variation remains substantial in both periods.
- The weaker 2013–2016 first stage does not appear to be driven by a lack of within-program variation.

Inframarginal Student Definitions

Identify students who would have enrolled in the target program regardless of small changes in annual admission capacity.

Construction

- For each program, compute the annual number of students who actually enrolled.
- Define an enrollment threshold using one of two alternative rules:
 - ▶ **First Cohort:** size of the first observed enrolled cohort.
 - ▶ **Minimum Cohort:** minimum enrolled cohort observed over the sample period.
- An applicant is classified as inframarginal if their admission rank is at or above the threshold defined for that program.

$$\text{Inframarginal}_{ipt} = 1 \iff \text{Rank}_{ipt} \leq T_p,$$

where T_p is either the first enrolled cohort size or the minimum enrolled cohort size for program p .

Results

	First Cohort	80 % First Cohort	Minimum Cohort	80 % Minimum Cohort
Infra students (N)	388,867	314,482	354,631	284,126
Share ranked infra (%)	72.90	58.96	66.48	53.27
FD observations	6,523	6,523	6,521	6,519
Programs	1,257	1,257	1,257	1,257
First stage	0.7317*** (0.0504)	0.7317*** (0.0504)	0.7317*** (0.0504)	0.7317*** (0.0504)
F-stat	210.7	210.7	210.6	210.6
2SLS Program	0.00009 (0.00022)	0.00011 (0.00023)	0.00018 (0.00027)	0.00019 (0.00029)

Notes: Columns differ only in the definition of inframarginal students. In all columns, the threshold used to define inframarginal status is based on effective enrollment. **First Cohort** defines inframarginal students as applicants whose admission rank is weakly below the number of students effectively enrolled in the first observed cohort of each program. **80 % First Cohort** restricts this threshold to 80 percent of the first enrolled cohort. **Minimum Cohort** uses the minimum annual effective enrollment observed for each program over the sample period. **80 % Minimum Cohort** restricts this minimum threshold to 80 percent. The first stage instruments annual changes in total program enrollment with annual changes in total vacancies. The outcome is graduation from the program in which the student initially enrolled within eight years. Robust standard errors clustered at the program level are reported in parentheses.

Alternative inframarginal

Definition	N	Share (%)	FD Obs.	First stage	F-stat	2SLS
<i>First Cohort</i>						
100 %	388,867	70.59	6,523	0.7317*** (0.0504)	210.7	0.00009 (0.00022)
90 %	350,913	63.70	6,523	0.7317*** (0.0504)	210.7	0.00011 (0.00023)
80 %	314,482	57.09	6,523	0.7317*** (0.0504)	210.7	0.00011 (0.00023)
70 %	276,422	50.18	6,521	0.7317*** (0.0504)	210.7	0.00009 (0.00023)
60 %	238,522	43.30	6,519	0.7317*** (0.0504)	210.7	0.00009 (0.00024)
<i>Minimum Cohort</i>						
100 %	354,631	64.37	6,521	0.7317*** (0.0504)	210.6	0.00018 (0.00027)
90 %	318,091	57.74	6,519	0.7317*** (0.0504)	210.6	0.00021 (0.00028)
80 %	284,126	51.58	6,519	0.7317*** (0.0504)	210.6	0.00019 (0.00029)
70 %	249,085	45.21	6,513	0.7317*** (0.0504)	210.6	0.00014 (0.00029)
60 %	214,682	38.97	6,511	0.7317*** (0.0504)	210.6	0.00020 (0.00030)

Notes: The table reports robustness checks using progressively more conservative definitions of inframarginal students. For each program, the baseline definition uses either the size of the first enrolled cohort (*First Cohort*) or the minimum enrolled cohort observed over 2007–2016 (*Minimum Cohort*). Alternative definitions restrict the inframarginal sample to the top 90 %, 80 %, 70 %, and 60 % of each baseline cohort size. All thresholds are based on effective enrollment and then applied to applicants according to their admission ranking. The first stage instruments annual changes in total enrollment with annual changes in total vacancies. The outcome is graduation from the initially enrolled program within eight years. Robust standard errors clustered at the program level are reported in parentheses.

Heterogeneity by University Selectivity: First Cohort

University	Mean PSU	First stage	F-stat	2SLS
<i>More selective universities</i>				
Univ. Católica	685.8	0.9925*** (0.0277)	1,284.7	-0.00023 (0.00036)
Univ. de Chile	675.7	0.8039*** (0.0448)	322.6	-0.00003 (0.00031)
Univ. de Santiago	616.7	0.8508*** (0.0315)	728.1	0.00113** (0.00053)
<i>Less selective universities</i>				
Univ. Playa Ancha	548.4	0.6235*** (0.0545)	131.1	-0.01385*** (0.00282)
Univ. Católica de Concepción	558.0	0.4448*** (0.0516)	74.3	0.00005 (0.00260)
Univ. del Bío-Bío	560.5	0.6681*** (0.1913)	12.2	-0.01598** (0.00797)

Notes: Results use the 2007–2016 panel and the *First Cohort* definition of inframarginal students. Mean PSU is the average Language–Mathematics PSU score of students enrolled in their target program. The first stage estimates the effect of annual changes in total vacancies on annual changes in total program enrollment. The 2SLS coefficient estimates the effect of one additional enrolled student induced by changes in vacancies on the eight-year graduation rate from the initially enrolled program among inframarginal students. Robust standard errors clustered at the program level are reported in parentheses. * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$.

Heterogeneity by University Selectivity: Minimum Cohort

University	Mean PSU	First stage	F-stat	2SLS
<i>More selective universities</i>				
Univ. Católica	685.8	0.9925*** (0.0277)	1,284.7	-0.00023 (0.00037)
Univ. de Chile	675.7	0.8039*** (0.0448)	322.6	-0.00008 (0.00031)
Univ. de Santiago	616.7	0.8508*** (0.0315)	728.1	0.00113** (0.00053)
<i>Less selective universities</i>				
Univ. Playa Ancha	548.4	0.6235*** (0.0545)	131.1	-0.01447*** (0.00293)
Univ. Católica de Concepción	558.0	0.4448*** (0.0516)	74.3	0.00007 (0.00285)
Univ. del Bío-Bío	560.5	0.6681*** (0.1913)	12.2	-0.01756** (0.00837)

Notes: Results use the 2007–2016 panel and the *Minimum Cohort* definition of inframarginal students. Mean PSU is the average Language–Mathematics PSU score of students enrolled in their target program. The first stage estimates the effect of annual changes in total vacancies on annual changes in total program enrollment. The 2SLS coefficient estimates the effect of one additional enrolled student induced by changes in vacancies on the eight-year graduation rate from the initially enrolled program among inframarginal students. Robust standard errors clustered at the program level are reported in parentheses. * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$.

Program selectivity and graduation effects

Panel A. First Cohort

Program selectivity	Programs	FD obs.	First Stage	F-stat	2SLS
PSU < 550	410	1,400	0.4356*** (0.0296)	217.0	0.00125 (0.00112)
PSU 550–600	467	2,563	0.5165*** (0.0369)	196.3	0.00036 (0.00071)
PSU 600–650	249	1,665	0.7510*** (0.0486)	238.9	-0.00032 (0.00053)
PSU ≥ 650	131	895	0.9229*** (0.0326)	802.3	-0.00009 (0.00018)

Panel B. Minimum Cohort

Program selectivity	Programs	FD obs.	First Stage	F-stat	2SLS
PSU < 550	410	1,400	0.4356*** (0.0296)	217.0	0.00140 (0.00117)
PSU 550–600	467	2,561	0.5162*** (0.0369)	196.0	0.00048 (0.00077)
PSU 600–650	249	1,665	0.7510*** (0.0486)	238.9	-0.00023 (0.00052)
PSU ≥ 650	131	895	0.9229*** (0.0326)	802.3	-0.00004 (0.00025)

Notes: Programs are classified according to the average Language–Mathematics PSU score of enrolled students over 2007–2016: PSU < 550, 550–600, 600–650, and PSU ≥ 650. Standard errors clustered at the program level are reported in parentheses. Outcomes correspond to graduation from the initially enrolled program within eight years.

Heterogeneity by Field of Study

Field group	Programs	FD obs.	First stage	F-stat	2SLS
Panel A. First Cohort					
Business and Law	130	651	0.6305*** (0.0654)	93.0	0.00179** (0.00084)
Health	221	1,290	0.4074*** (0.0342)	142.1	-0.00013 (0.00082)
STEM	432	2,162	0.8502*** (0.0475)	320.7	-0.00015 (0.00010)
Education and Humanities	372	1,913	0.7158*** (0.0994)	51.9	-0.00060 (0.00061)
Arts, Architecture and Agriculture	102	507	0.6541*** (0.0557)	138.0	0.00070 (0.00120)
Panel B. Minimum Cohort					
Business and Law	130	651	0.6305*** (0.0654)	93.0	0.00189** (0.00084)
Health	221	1,288	0.4069*** (0.0342)	141.9	0.00013 (0.00084)
STEM	432	2,162	0.8502*** (0.0475)	320.7	-0.00014 (0.00012)
Education and Humanities	372	1,913	0.7158*** (0.0994)	51.9	-0.00062 (0.00064)
Arts, Architecture and Agriculture	102	507	0.6541*** (0.0557)	138.0	0.00073 (0.00117)

Notes: Results use the 2007–2016 panel. Fields are grouped using the target program's field of study and fixed at the program level. The first stage estimates the effect of annual changes in vacancies on annual changes in total enrollment. The 2SLS coefficient estimates the effect of one additional enrolled student, induced by changes in vacancies, on the eight-year graduation rate from the initially enrolled program among inframarginal students. Robust standard errors clustered at the program level are reported in parentheses. * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$.

Inframarginal Students: Graduation at 8 vs. 10 Years

Definition	Horizon	Programs	FD obs.	First stage	2SLS
First Cohort	8 years	1,092	4,881	0.7413*** (0.0579)	0.00002 (0.00021)
First Cohort	10 years	1,092	4,881	0.7413*** (0.0579)	0.00031 (0.00034)
Minimum Cohort	8 years	1,092	4,881	0.7413*** (0.0579)	0.00010 (0.00027)
Minimum Cohort	10 years	1,092	4,881	0.7413*** (0.0579)	0.00048 (0.00043)

Notes: The table compares graduation outcomes measured within eight and ten years using the same 2007–2014 sample. The outcome is graduation from the program in which the student initially enrolled. Each estimate uses 1,092 programs and 4,881 first-difference observations. The first stage instruments annual changes in total program enrollment with annual changes in total vacancies; the first-stage coefficient implies that 10 additional vacancies increase enrollment by approximately 7.4 students. Robust standard errors clustered at the program level are reported in parentheses. * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$.

Inframarginal IV results

Outcome	First stage	F-stat	Reduced form	2SLS
Target graduation, 8y	0.732 (0.050)	210.7	0.00013 (0.00020)	0.00019 (0.00028)
University graduation, 8y	0.732 (0.050)	210.7	0.00005 (0.00012)	0.00007 (0.00016)
Higher education graduation, 8y	0.732 (0.050)	210.7	0.00005 (0.00012)	0.00007 (0.00016)

- 494,675 inframarginal students (90.6 % of enrollment).

Heterogeneity by university selectivity

University	Mean PSU	First stage	F-stat	Reduced form	2SLS
<i>Highly selective universities</i>					
UC	693.1	0.993 (0.028)	903.9	0.00024 (0.00028)	0.00024 (0.00028)
UCH	686.1	0.804 (0.045)	390.5	0.00096*** (0.00026)	0.00119*** (0.00036)
<i>Less selective universities</i>					
UNAB	559.2	0.380 (0.040)	90.8	-0.00029 (0.00022)	-0.00076 (0.00060)
UCSC	563.5	0.445 (0.052)	75.4	-0.00174** (0.00076)	-0.00385** (0.00157)
UBB	567.5	0.668 (0.191)	14.1	-0.00464** (0.00206)	-0.00673* (0.00366)

- Mean PSU is based on target-program selectivity.
- Outcome: university graduation among inframarginal students.
- Negative effects emerge mainly among less selective institutions.

Entry of private universities into the SUA

Research question

Did first-year enrollment in incumbent CRUCH programs change after eight private universities entered the SUA in 2012, particularly in programs with greater pre-existing exposure to these entrants?

Sample

- Incumbent CRUCH program $\times$ admission-year panel, 2007–2016.
- Outcome: number of students entering the first year of program p in year t .

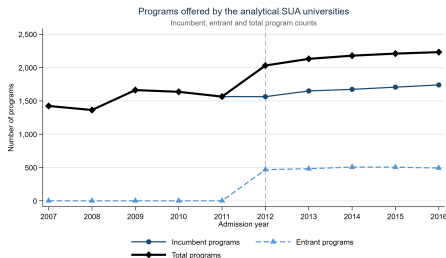
Exposure design

- Markets are defined by pre-treatment region $\times$ academic field.
- Exposure measures the 2009–2011 enrollment presence of the future entrant universities in the incumbent program's market.
- Total exposure uses the entrant enrollment share; Triangular and Gaussian measures additionally weight entrants by PSU-score proximity.
- Exposure is fixed over time and interacted with the post-entry period, 2012–2016.

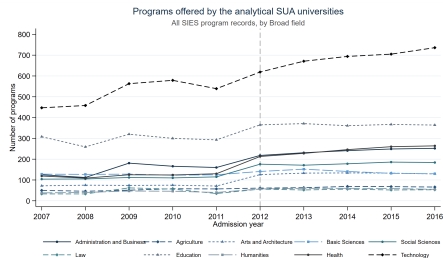
The design compares pre/post-2012 enrollment changes across incumbent programs with different levels of exposure to the future entrants.

Evolution of the number of SUA programs

Programs by university status



Programs by Broad field



- The incorporation of the eight entrant universities adds roughly 470 programs in 2012. Their program supply remains relatively stable afterward, while incumbents account for most subsequent growth.
- Technology represents the largest field and an important share of program expansion.

Construction of the exposure measures

1. Pre-reform enrollment stocks in market m

$$A_m = \sum_{\substack{q \in \mathcal{E} \\ m(q)=m}} \bar{N}_{q,2009-2011}^{\text{first}}, \quad T_m = \sum_{\substack{q \\ m(q)=m}} \bar{N}_{q,2009-2011}^{\text{first}}.$$

- $\mathcal{E}$ is the set of entrant programs and $m(q)$ is program q 's pre-treatment region $\times$ field market.
- $\bar{N}_{q,2009-2011}^{\text{first}}$ is average first-year enrollment; hence A_m is entrant enrollment and T_m total enrollment in market m .

2. Total exposure of incumbent program p

$$E_p = \frac{A_{m(p)}}{T_{m(p)}}.$$

Total exposure is the entrant share of pre-reform first-year enrollment in the market of incumbent program p . Thus, incumbents in the same pre-treatment region $\times$ field market share the same E_p .

3. Selectivity-weighted exposure of incumbent program p

$$E_p^k = \frac{\sum_{\substack{q \in \mathcal{E} \\ m(q)=m(p)}} k(|S_p - S_q|) \bar{N}_{q,2009-2011}^{\text{first}}}{T_{m(p)}}, \quad k \in \{\text{Triangular, Gaussian}\}.$$

- S_p and S_q are pre-reform mean PSU scores of incumbent p and entrant q ; $|S_p - S_q|$ is their selectivity distance.
- $k(\cdot)$ gives greater weight to closer entrants, so E_p^k can vary across incumbent programs within the same market.

SUA exposure: samples and descriptive statistics

Panel A: Analytical samples

Variable	Broad	ISCED-97	Generic
Programs	949	949	945
Program-year observations	8,041	8,030	8,002
Incumbent universities	25	25	25
Pre-treatment markets	130	215	638
Mean annual enrollment, 2009–2011	53.2	53.2	53.2
Mean PSU score, 2009–2011	588.8	588.8	589.0

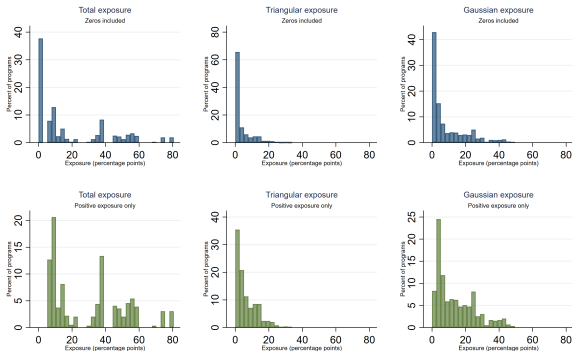
Panel B. Exposure by academic-field definition

Field definition	Exposure measure	Mean	Std. dev.	Share zero
Broad	Total	18.53	22.21	37.7 %
	Triangular	3.48	5.57	46.8 %
	Gaussian	8.58	11.26	37.7 %
ISCED-97	Total	17.88	22.96	43.0 %
	Triangular	3.16	5.79	56.9 %
	Gaussian	8.20	11.70	43.0 %
Generic	Total	13.27	23.40	66.2 %
	Triangular	1.83	6.13	83.4 %
	Gaussian	5.83	11.90	67.4 %

Notes: Panel A reports the analytical sample under each field definition. Enrollment is the unweighted mean across programs of average annual first-year enrollment in 2009–2011; PSU is the program's mean score over the same period. Panel B reports unweighted exposure moments in percentage points, including zero-exposure programs. Total is the entrant share of market enrollment; Triangular and Gaussian additionally weight entrant enrollment by PSU proximity using a 50-point bandwidth. Markets combine pre-treatment region with the corresponding field definition.

SUA exposure distributions: Broad field

One observation per incumbent program



Top row includes zeros; bottom row conditions on positive exposure.

- Each incumbent program appears once. The horizontal axis reports exposure in percentage points and the vertical axis the share of programs.
- The upper row includes all programs; the lower row excludes zeros and shows the distribution conditional on positive exposure.
- Total exposure has the largest values. Triangular weighting is the most restrictive, while Gaussian weighting produces an intermediate distribution.

First-stage specification

$$N_{pt}^{\text{first}} = \pi_k (E_p^k \times \text{Post}_t) + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \nu_{pt}.$$

- p indexes incumbent programs and t years. N_{pt}^{first} is first-year enrollment of program p in year t , while E_p^k is its pre-reform exposure and $\text{Post}_t = 1$ in 2012–2016.
- $f(p)$ denotes the pre-treatment field of program p and $r(p)$ its pre-treatment region. Accordingly, $\alpha_{f(p),t}$ are field $\times$ year fixed effects and $\gamma_{r(p),t}$ are region $\times$ year fixed effects.
- μ_p denotes program fixed effects, and π_k is the post-2012 enrollment difference associated with a 10 percentage-point increase in exposure measure k .

Exposure is positive in only four regions

- In the 2009–2011 analytical sample, entrant universities offered regular undergraduate programs only in Metropolitana, Valparaíso, Biobío and La Araucanía.
- Thus, these are the only regions containing entrant programs and positive entrant enrollment before integration into the SUA.
- In every other region, the entrant-enrollment numerator is zero: $A_m = 0$, and therefore $E_p = A_{m(p)} / T_{m(p)} = 0$.

$$N_{pt}^{\text{first}} = \pi \left(E_p^{k,m} \times \text{Post}_t \right) + \mu_p + \alpha_{f_m(p),t} + \gamma_{r(p),t} + \nu_{pt}.$$

Exposure measure	Field × year FE			+ Region × year FE		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>Total</i>	1.967*** (0.336)	1.893*** (0.335)	1.872*** (0.289)	0.798 (0.662)	0.904* (0.510)	0.324 (0.338)
Wald <i>F</i>	34.36	31.92	41.90	1.45	3.14	0.92
<i>Triangular</i>	0.935 (0.972)	1.392 (0.934)	2.185** (0.927)	-5.252** (2.115)	-3.415*** (1.310)	-0.975 (0.829)
Wald <i>F</i>	0.93	2.22	5.56	6.17	6.80	1.39
<i>Gaussian</i>	1.775*** (0.439)	1.939*** (0.450)	2.342*** (0.533)	-2.767* (1.401)	-1.447* (0.832)	-0.558 (0.571)
Wald <i>F</i>	16.36	18.60	19.33	3.90	3.03	0.96
Observations	8,041	8,030	7,756	8,038	8,027	7,753
Programs	949	949	922	948	948	921
Markets	130	215	615	129	214	614
Program FE	Yes	Yes	Yes	Yes	Yes	Yes
Field × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × year FE	No	No	No	Yes	Yes	Yes

Notes: The dependent variable is first-year enrollment in incumbent CRUCH programs. Total exposure is entrants' pre-treatment enrollment share; Triangular and Gaussian additionally weight entrant enrollment by PSU-score proximity using a 50-point bandwidth. Markets combine pre-treatment region with the indicated field definition. Exposure is interacted with Post_t , equal to one in 2012–2016, and measured in 10 percentage-point units. All regressions absorb program and corresponding field × year fixed effects; the last three columns additionally absorb region × year fixed effects. Standard errors are clustered by the corresponding pre-treatment market and reported in parentheses. Wald *F* statistics test the exposure coefficient. Sample differences reflect singleton removal by `reghdfe`. *** $p < 0,01$, ** $p < 0,05$, * $p < 0,10$.

$$\log N_{pt}^{\text{first}} = \beta_k \left[\widetilde{\log E_p^k} \times \text{Post}_t \right] + \theta_k \left[D_p^k \times \text{Post}_t \right] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt},$$

$$\widetilde{\log E_p^k} = \begin{cases} \log(E_p^k), & E_p^k > 0, \\ 0, & E_p^k = 0, \end{cases} \quad D_p^k = \mathbf{1}\{E_p^k > 0\}.$$

Exposure measure	Field × year FE			+ Region × year FE		
	Broad	ISCED-97	Generic	Broad	ISCED-97	Generic
<i>Total</i>	0.100*** (0.034)	0.072** (0.035)	0.021 (0.030)	−0.005 (0.039)	−0.024 (0.032)	−0.022 (0.027)
<i>F</i>	8.84	4.24	0.48	0.01	0.55	0.69
<i>Triangular</i>	−0.013 (0.015)	0.004 (0.016)	0.016 (0.021)	−0.050*** (0.017)	−0.024 (0.016)	−0.005 (0.017)
<i>F</i>	0.81	0.07	0.59	8.95	2.37	0.07
<i>Gaussian</i>	−0.026 (0.032)	−0.024 (0.030)	0.001 (0.018)	−0.107*** (0.030)	−0.079*** (0.026)	−0.025 (0.017)
<i>F</i>	0.68	0.64	0.01	12.72	9.35	2.19
Observations	8,041	8,030	7,756	8,038	8,027	7,753
Programs	949	949	922	948	948	921

Notes: Each coefficient reports β_k , the post-2012 elasticity with respect to exposure intensity. Zero-exposure programs are retained by setting $\widetilde{\log E_p^k} = 0$ when $E_p^k = 0$ and including $D_p^k \times \text{Post}_t$ separately; the coefficient on this indicator is not reported. Thus, β_k captures variation in exposure intensity among positively exposed programs while allowing zero-exposure programs to remain in the estimation sample. All specifications absorb program and corresponding field × year fixed effects; the last three columns additionally absorb pre-treatment region × year fixed effects. Standard errors are clustered at the pre-treatment market level. Observations with zero first-year enrollment are excluded because the dependent variable is in logs. * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$.

Decomposing weighted exposure

$$E_p^k = \underbrace{\frac{\sum_{q \in \mathcal{E}_{m(p)}} k(|S_p - S_q|) N_q}{\sum_{\ell \in m(p)} N_\ell}}_{\text{Direct construction of weighted exposure}} \equiv \underbrace{\left[\frac{\sum_{q \in \mathcal{E}_{m(p)}} N_q}{\sum_{\ell \in m(p)} N_\ell} \right]}_{M_{m(p)} \text{ entrant market share}} \times \underbrace{\left[\frac{\sum_{q \in \mathcal{E}_{m(p)}} k(|S_p - S_q|) N_q}{\sum_{q \in \mathcal{E}_{m(p)}} N_q} \right]}_{Q_p^k \text{ similarity with entrants}}$$

Entrant presence

$$M_m = \frac{\sum_{q \in \mathcal{E}_m} N_q}{\sum_{\ell \in m} N_\ell}.$$

M_m is the entrant share of total market enrollment. It captures the **size of entrant presence** and is common to all incumbent programs in market m .

Conditional similarity

$$Q_p^k = \frac{\sum_{q \in \mathcal{E}_m} k(|S_p - S_q|) N_q}{\sum_{q \in \mathcal{E}_m} N_q}.$$

Q_p^k is the entrant-enrollment-weighted PSU similarity between incumbent p and the entrant programs in its market. It is higher when larger entrants have PSU profiles closer to that of p . Conditional on entrant presence, it captures **competitive proximity** rather than the overall size of entry.

Using similarity as the independent variable

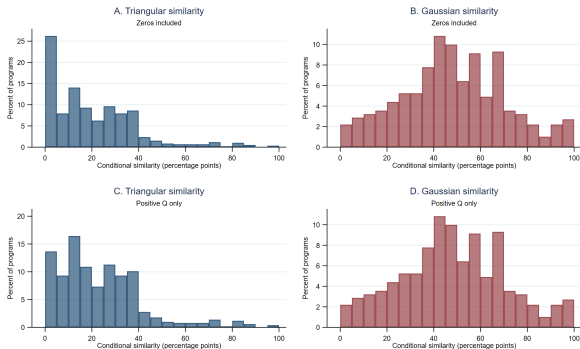
$$N_{pt}^{\text{first}} = \beta_k \left[Q_p^k \times \text{Post}_t \right] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}, \quad M_{m(p)} > 0.$$

The sample is restricted to markets with positive entrant presence. Program fixed effects absorb time-invariant differences across incumbents, while Broad-field $\times$ year and pre-treatment region $\times$ year fixed effects absorb common field-specific and regional shocks over time. The model therefore asks whether, among incumbents in exposed markets, programs whose pre-treatment PSU profiles are more similar to the entrants experience different post-2012 enrollment changes. Unlike E_p^k , which combines entrant market share and similarity, Q_p^k isolates the **similarity component** of exposure. The coefficient β_k measures the differential post-2012 enrollment change associated with a 0.10 increase in entrant-enrollment-weighted PSU similarity.

Distribution of conditional similarity measures

Distribution of conditional similarity measures

One observation per incumbent program in exposed markets



The upper row includes zeros; the lower row conditions on positive Q .
The horizontal axis reports conditional similarity in percentage points and the vertical axis reports the share of programs.

- Each incumbent program appears once. The horizontal axis reports conditional similarity Q in percentage points and the vertical axis the share of programs.
- The upper row includes all exposed programs, while the lower row excludes zeros and shows the distribution conditional on positive Q .
- Triangular similarity is concentrated at relatively low values and contains a relevant mass near zero. Gaussian similarity is more dispersed and shifted toward higher values, indicating a less restrictive measure of similarity across exposed markets.

Common fixed-effects specification

$$N_{pt}^{\text{first}} = \beta_k [Q_p^k \times \text{Post}_t] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}, \quad \log N_{pt}^{\text{first}} = \delta_k [\widetilde{\log Q_p^k} \times \text{Post}_t] + \theta_k [D_p^k \times \text{Post}_t] + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt},$$

$$\widetilde{\log Q_p^k} = \begin{cases} \log(Q_p^k), & Q_p^k > 0, \\ 0, & Q_p^k = 0, \end{cases} \quad D_p^k = \mathbf{1}\{Q_p^k > 0\}, \quad M_m > 0.$$

	Levels		Logs	
	Triangular	Gaussian	Triangular	Gaussian
Similarity term $\times$ Post	-2,381*** (0,491)	-2,241*** (0,487)	-0,067*** (0,020)	-0,142*** (0,035)
Wald F	23.50	21.18	11.37	16.29
Observations	5,009	5,009	5,009	5,009
Programs	591	591	591	591
Markets	36	36	36	36

Notes: In levels, a 0.10 increase in conditional similarity is associated with approximately 2.2–2.4 fewer first-year students after 2012. In logs, a 10 % increase in similarity is associated with approximately 0.67 % lower enrollment under the triangular measure and 1.42 % lower enrollment under the Gaussian measure. The sample is restricted to markets with positive entrant presence, $M_m > 0$. Levels retain economically meaningful observations with $Q_p^k = 0$. In logs, zero-

similarity observations are also retained by setting $\widetilde{\log Q_p^k} = 0$ when $Q_p^k = 0$ and including $D_p^k \times \text{Post}_t$ separately. The coefficient on this indicator is not reported. Gaussian similarity is strictly positive in the exposed-market sample, so the corresponding indicator is absorbed by the fixed effects. All regressions absorb program, Broad-field $\times$ year, and pre-treatment region $\times$ year fixed effects. Standard errors clustered by pre-treatment market are in parentheses. Event studies reject the absence of differential pre-treatment trends, so estimates are interpreted as descriptive rather than causal.

*** $p < 0,01$.

First-stage heterogeneity by incumbent selectivity

	PSU < 550		PSU 550–649		PSU ≥ 650	
	Tri.	Gau.	Tri.	Gau.	Tri.	Gau.
<i>SUA exposure</i>						
Coefficient	−1,61 (2,78)	−0,54 (2,10)	−3,17** (1,54)	−0,98 (0,83)	−15,75* (8,25)	−5,01 (4,38)
F-statistic	0.34	0.07	4.26	1.41	3.65	1.31
Observations	2,215		4,609		1,188	
Programs	272		531		143	
Clusters	89		100		25	
<i>Conditional similarity Q</i>						
Coefficient	−1,95* (0,93)	−3,26 (1,96)	−2,20*** (0,48)	−2,17*** (0,37)	−5,70 (4,24)	−3,48 (3,15)
F-statistic	4.40	2.77	21.27	34.14	1.80	1.22
Observations	836		2,932		1,038	
Programs	104		339		128	
Clusters	15		28		16	

Notes: Standard errors clustered by pre-treatment market in parentheses. Panel A includes program, Broad-field × year, and region × year FE. Panel B includes program and market × year FE and is restricted to exposed markets. Coefficients correspond to a 10 percentage-point increase in exposure or a 0.10 increase in conditional similarity. * $p < 0,10$, ** $p < 0,05$, *** $p < 0,01$.

Cosine exposure

Each program is represented by a vector describing the distribution of its 2011 first-year students. Each coordinate counts students with a particular PSU profile and, depending on the definition, campus region and Broad field. Programs are more similar when their students are distributed across the same vector cells.

1. Program vectors

$$\mathbf{v}_p^{PSU} = (v_{p,b})_{b \in \mathcal{B}}, \quad \mathbf{v}_p^{PSU \times R} = (v_{p,rb})_{\substack{r \in \mathcal{R}, \\ b \in \mathcal{B}}}, \quad \mathbf{v}_p^{PSU \times R \times F} = (v_{p,rfb})_{\substack{r \in \mathcal{R}, f \in \mathcal{F}, \\ b \in \mathcal{B}}}.$$

Here, b denotes a 25-point PSU bin, r campus region, and f Broad field. Each vector cell contains the number of first-year students in that combination. Entrant-program vectors are constructed analogously.

2. Similarity between incumbent p and entrant j

$$\cos \theta_{j,p}^d = \frac{\mathbf{v}_j^d \cdot \mathbf{v}_p^d}{\|\mathbf{v}_j^d\| \|\mathbf{v}_p^d\|}, \quad d \in \{PSU, PSU \times R, PSU \times R \times F\}.$$

The cosine compares the **shape of the student distributions**, rather than program size. Under interaction-cell definitions, similarity is zero when programs do not share the indicated region or field.

3. Additive similarity measures

$$\cos \theta_{j,p}^{PSU+R} = \frac{\cos \theta_{j,p}^{PSU} + \mathbf{1}\{r_j = r_p\}}{2}, \quad \cos \theta_{j,p}^{PSU+R+F} = \frac{\cos \theta_{j,p}^{PSU} + \mathbf{1}\{r_j = r_p\} + \mathbf{1}\{f_j = f_p\}}{3}.$$

These measures give equal weight to PSU-profile similarity and coincidence in campus region and, where applicable, Broad field.

4. Exposure of incumbent program p

$$E_p^d = \sum_{j \in \mathcal{J}} w_j \cos \theta_{j,p}^d, \quad w_j = \frac{N_{j,2011}^{\text{first}}}{\sum_{\ell \in \mathcal{J}} N_{\ell,2011}^{\text{first}}}.$$

Exposure is higher when incumbent p resembles entrant programs that account for a larger share of entrant enrollment.

Notes: $\mathcal{J}$ contains programs from the eight entrant universities. Campus region is the university program's location in 2011. Student school region and SES are not used.

Cosine exposure: samples and descriptive statistics

Panel A: Analytical samples

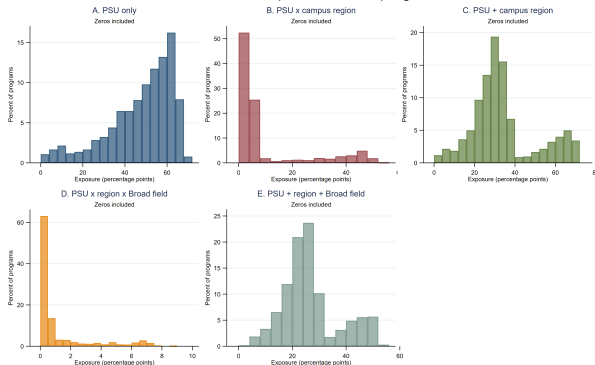
Variable	Full sample	$N_{p,2011}^{\text{first}} \geq 10$
Programs	996	939
Program-year observations	9,105	8,618
Incumbent universities	25	25
Total first-year enrollment, 2011	54,659	54,303
Mean enrollment per program, 2011	54.9	57.8
Mean annual enrollment, 2007–2011	54.8	57.3
Mean PSU score, 2011	578.0	580.9

Panel B: Distribution of raw cosine exposure

Exposure measure	Full sample			$N_{p,2011}^{\text{first}} \geq 10$		
	Mean	Std. dev.	Share zero	Mean	Std. dev.	Share zero
PSU bins	0.4762	0.1559	0.0 %	0.4842	0.1540	0.0 %
PSU bins $\times$ campus region	0.0977	0.1495	37.0 %	0.1024	0.1521	34.7 %
PSU bins + campus region	0.3405	0.1614	0.0 %	0.3490	0.1604	0.0 %
PSU bins $\times$ campus region $\times$ Broad field	0.0113	0.0196	38.3 %	0.0119	0.0200	35.8 %
PSU bins + campus region + Broad field	0.2681	0.1078	0.0 %	0.2745	0.1064	0.0 %

Notes: The unit is an incumbent SUA program and exposure moments are unweighted across programs. The restricted sample excludes 57 programs with fewer than 10 first-year students in 2011. Multiplicative measures require coincidence in the indicated dimensions; additive measures assign equal weight to PSU similarity, campus-region coincidence and, where applicable, Broad-field coincidence. The restriction is applied after exposure construction, so entrant profiles, weights, and exposure scaling remain unchanged. Campus region is the university program's location; student school region and SES are not used.

Cosine exposure distributions One observation per incumbent program



Zeros included. Exposure measured in percentage points; each panel uses its own x-axis scale.

- Each incumbent program appears once. The horizontal axis reports cosine exposure in percentage points and the vertical axis the share of programs.
- Multiplicative definitions are more restrictive. In particular, $\text{PSU} \times \text{campus region} \times \text{Broad field}$ places substantial mass at zero and has a much smaller support than the PSU-only and additive measures.

$$N_{pt}^{\text{first}} = \beta_k(E_p^k \times \text{Post}_t) + \mu_p + \alpha_{f(p),t} + \gamma_{r(p),t} + \varepsilon_{pt}.$$

Exposure measure	Full sample			$N_{p,2011}^{\text{first}} \geq 10$		
	(1) + year	(2) + field × year	(3) + region × year	(4) + year	(5) + field × year	(6) + region × year
PSU bins	−1,811* (0,948)	−1,403 (1,018)	−0,716 (1,002)	−2,116** (1,017)	−1,597 (1,098)	−0,630 (1,075)
Wald F	3,65	1,90	0,51	4,33	2,11	0,34
PSU bins × region	3,976*** (0,601)	4,126*** (0,598)	−5,174 (3,736)	3,903*** (0,611)	4,119*** (0,611)	−5,267 (3,797)
Wald F	43,82	47,55	1,92	40,82	45,41	1,92
PSU bins × region × field	3,680*** (0,658)	3,624*** (0,647)	−3,922* (2,075)	3,590*** (0,666)	3,612*** (0,655)	−4,023* (2,140)
Wald F	31,33	31,36	3,57	29,09	30,44	3,53
PSU bins + region	4,299*** (0,643)	4,416*** (0,641)	−1,476 (2,066)	4,344*** (0,667)	4,577*** (0,668)	−1,298 (2,217)
Wald F	44,76	47,54	0,51	42,36	47,02	0,34
PSU bins + region + field	4,639*** (0,656)	4,429*** (0,642)	−1,480 (2,071)	4,728*** (0,690)	4,590*** (0,669)	−1,302 (2,224)
Wald F	50,07	47,54	0,51	46,97	47,02	0,34
Observations	9,105	9,105	9,103	8,618	8,618	8,610
Programs	996	996	996	939	939	938

Notes: The dependent variable is incumbent-program first-year enrollment. Coefficients are reported first, program-clustered standard errors are in parentheses, and Wald F statistics appear in the corresponding labeled row. Exposure is measured in SD units and $\text{Post}_t = 1$ in 2012–2016. Columns (1) and (4) include program and year FE; Columns (2) and (5) use Broad-field × year FE; Columns (3) and (6) add campus-region × year FE. Multiplicative measures require coincidence in the indicated dimensions; additive measures combine them with equal weights. Columns (4)–(6) exclude programs with fewer than 10 first-year students in 2011. *** $p < 0,01$, ** $p < 0,05$, * $p < 0,10$.

Design III: sudden vacancy expansions at competing programs

Shock at program j

- Regular-admission vacancies V_{jt} from DEMRE's *Oferta Académica* (first-semester vacancies), the seats filled in the centralized process.
- $\Delta \log V_{jt}$ in the top 20 % of positive changes (pooled 2008–2016), and *sudden*: j existed two years before, was not a top-20 % change in $t - 1$, and the increase is not reversed in $t + 1$.
- Shocks dated 2010–2016; SUA entrants are never shock sources.

Exposure of incumbent k

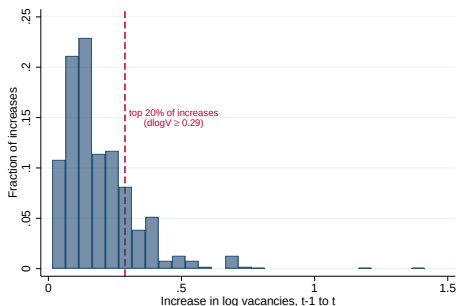
$$E_{kt} = \sum_j s(k \leftarrow j) \frac{(V_{jt} - V_{j,t-1}) \text{shock}_{jt}}{\bar{N}_k^{2007-09}}$$

Rank-order-list similarity (2007–2009 applicant lists, predetermined)

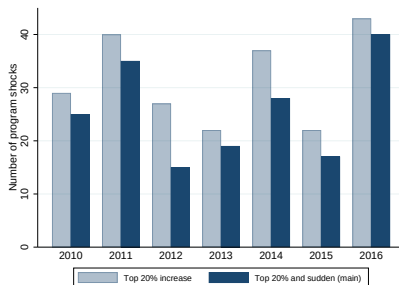
$$s(k \leftarrow j) = \frac{\sum_i \mathbf{1}\{j, k \in L_i\} / |r_{ij} - r_{ik}|}{\#\{i : k \in L_i\}}$$

Directed: normalized by k 's applicant pool, so a large j can press on a small k without the reverse. Variants: adjacent ranks only, both in the top two, plain co-listing; benchmark: same field $\times$ region.

Design III: distribution of vacancy changes



Program-years 2010-2016 with vacancies in t-1 and t: N = 6876. 71% have no change, 15% increase (1008, shown; trimmed at 1.5). Threshold: 80th percentile of positive changes, pooled 2008-2016.



- 71 % of program-years have no change in vacancies and 15 % increase. The top-20 % threshold is $\Delta \log V \geq 0.29$ ($= \log 4/3$: vacancies are round numbers, so many increases sit exactly at +33 %).
- Shocks are rare and mostly one-off (next slide). They cluster in 2011, 2014 and 2016.

Design III: where are the shocks?

	Program-years	Shocks	Shock rate (%)	Share of shocks (%)
<i>Pre-period selectivity (entrant PSU quintile)</i>				
No PSU data	809	20	2.5	11.2
Q1 (least selective)	976	31	3.2	17.3
Q2	1,150	26	2.3	14.5
Q3	1,288	29	2.3	16.2
Q4	1,331	40	3.0	22.3
Q5 (most selective)	1,322	33	2.5	18.4
<i>Field of study</i>				
Tecnología	2,015	55	2.7	30.7
Educación	1,441	36	2.5	20.1
Administración y Comercio	509	19	3.7	10.6
Salud	886	19	2.1	10.6
Ciencias Básicas	565	14	2.5	7.8
Arte y Arquitectura	347	13	3.7	7.3
Ciencias Sociales	625	12	1.9	6.7
Humanidades	157	10	6.4	5.6
Derecho	142	1	0.7	0.6
Agropecuaria	189	0	0.0	0.0
<i>University</i>				
Private (CRUCH)	2,563	59	2.3	33.0
State (CRUCH)	4,313	120	2.8	67.0
<i>Year</i>				
2010	929	25	2.7	14.0
2011	936	35	3.7	19.6
2012	946	15	1.6	8.4
2013	980	19	1.9	10.6
2014	1,026	28	2.7	15.6
2015	999	17	1.7	9.5
2016	1,060	40	3.8	22.3
Total	6,876	179	2.6	100.0

Shocks per program, 2010–2016	Programs
0	1,196
1	149
2	15
Total	1,360

- Shocks are spread across the selectivity distribution; no strong concentration in selective programs.
- More frequent at state universities and in Humanities and Business; almost absent in Law and Agriculture.
- Before 2012 only CRUCH universities were in DEMRE, so all incumbents are CRUCH.

Design III: shock definitions

Definition		Shocks	Mean $\Delta \log V$	Mean seats added
Main	top 20 %, sudden	179	0.39	18.5
Top 10 %	sudden	79	0.50	24.4
Top 25 %	sudden	205	0.37	18.8
Within-year p80	sudden	188	0.38	18.3
No sudden conditions	top 20 %	220	0.39	18.5
Level change	top 20 % of ΔV	152	0.35	25.4
INDICES vacancies	top 20 %, sudden	127	0.40	21.1
Admissions as vacancies	top 20 %, sudden	319	0.52	16.8

- The “sudden” conditions remove about a fifth of top-20 % increases.
- INDICES vacancies yield fewer shocks because only part of the programs link through SIES codes; regular admissions (status 24) as the vacancy measure yield more.

Design III: TWFE on cumulative ROL exposure

	Log enrollment		Enrollment		Entrant PSU		Cutoff	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cumulative ROL exposure (1 SD)	0.036** (0.014)	0.017 (0.016)	1.344*** (0.384)	-0.152 (0.464)	-0.962* (0.516)	-0.817* (0.443)	0.186 (0.579)	0.709 (0.580)
Own vacancy shocks		0.238*** (0.034)		12.921*** (1.560)		-3.712*** (1.175)		-7.359*** (2.388)
Mean of outcome	3.72	3.72	53.42	53.42	584.49	584.49	560.18	560.18
Field $\times$ year, region $\times$ year FE	No	Yes	No	Yes	No	Yes	No	Yes
Own-shock control	No	Yes	No	Yes	No	Yes	No	Yes
Observations	8,865	8,865	8,915	8,915	8,865	8,865	8,915	8,915

- $y_{kt} = \beta \text{cumE}_{kt} + \delta \text{own shocks}_{kt} + \mu_k + \alpha_{f(k)t} + \gamma_{r(k)t} + \varepsilon_{kt}$; exposure standardized; SE clustered by program.
- No effect on enrollment once field- and region-year shocks are absorbed. Entrant PSU falls with exposure (significant at 10%).
- Own vacancy shocks move own enrollment strongly, so they are controlled for.

Design III: robustness of the TWFE estimate

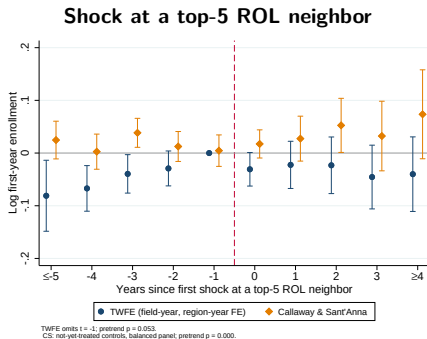
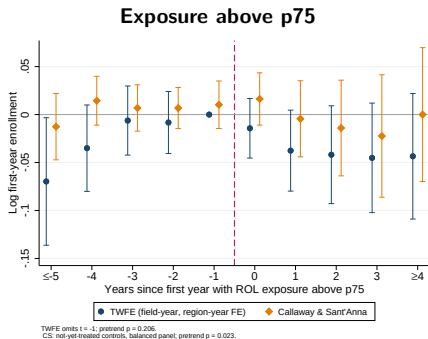
Shock definitions (log enrollment)

	Main (1)	Top 10 % (2)	Top 25 % (3)	Within-year (4)	No sudden (5)	Levels (6)	INDICES (7)	Admissions (8)
Cumulative ROL exposure (1 SD)	0.017 (0.016)	0.019 (0.014)	0.024** (0.011)	0.015 (0.017)	0.012 (0.023)	0.036*** (0.009)	0.001 (0.009)	0.005 (0.015)
Observations	8,865	8,865	8,865	8,865	8,865	8,865	8,865	8,865

Similarity measures (log enrollment)

	Rank-order-list similarity				Market
	Inverse distance (1)	Adjacent (2)	Top two (3)	Co-listing (4)	Field × region (5)
Cumulative exposure (1 SD)	0.017 (0.016)	0.017 (0.014)	0.013* (0.008)	0.015 (0.018)	0.004 (0.023)
Observations	8,865	8,865	8,865	8,865	8,865

Design III: event study, log enrollment



- Treatment: first year of high exposure; never-treated programs as controls.
- Callaway & Sant'Anna uses not-yet-treated controls on the balanced panel and cannot absorb field- or region-year effects.

Design III: pre-trend tests

Treatment	Outcome	TWFE p	Callaway–Sant’Anna p
Exposure > p75	Log enrollment	0.206	0.023
	Entrant PSU	0.129	0.029
	Cutoff	0.002	0.000
Exposure > p90	Log enrollment	0.974	0.000
	Entrant PSU	0.652	0.009
	Cutoff	0.004	0.000
Top-5 neighbor shock	Log enrollment	0.053	0.000
	Entrant PSU	0.061	0.000
	Cutoff	0.156	0.000

- Joint test of leads ≤ -5 to -2 . TWFE does not reject for log enrollment, but Callaway & Sant’Anna rejects in every specification, and TWFE rejects for the cutoff.
- As specified, Design III is not identified: programs exposed to competitor expansions were already on different paths.

Non-harmonized field observations

Institution	Program	Years	Code	Obs.
UV	Bachillerato ingreso común	2007–2008	1900	558
USACH	Tecnólogo en alimentos (vesp)	2007–2010	1690	208
UV	Administración pública – diurno	2007	1997	26
UV	Actuación teatral	2008	1928	9
Total		2007–2010		801

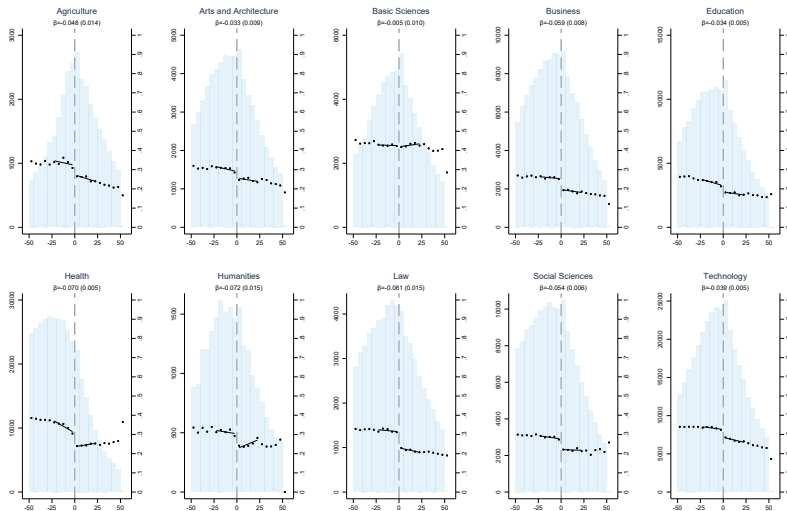
These observations are excluded from the field-enriched sample because their program field could not be reliably harmonized. They represent 0.095 % of the baseline RDD sample within the main bandwidth.

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Appendix note 2: PSU retakes by field

RDD: Retakes PSU Within Two Years by Field

Outcome: retakes_psu_2y. Regression BW = ± 25 ; plot range = ± 50

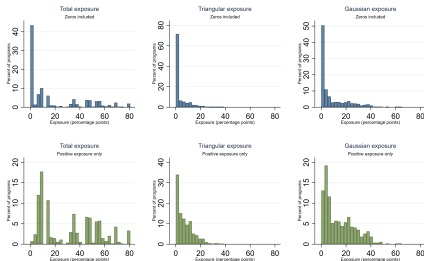


The reduction in PSU retaking is generally negative across fields, although magnitudes vary by academic area. [← Back](#)

SUA exposure: alternative field definitions

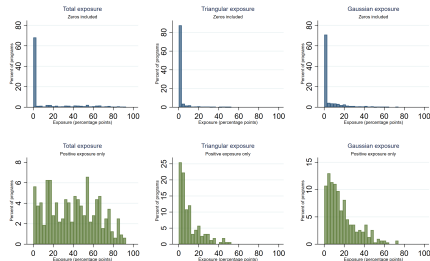
← Broad field

SUA exposure distributions: ISCED-97 field
One observation per incumbent program



Top row includes zero; bottom row conditions on positive exposure.

SUA exposure distributions: Generic field
One observation per incumbent program

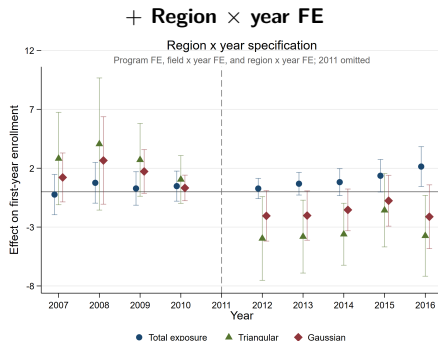
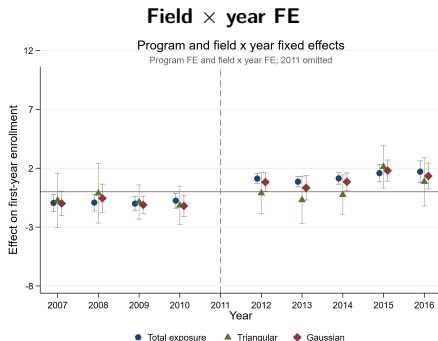


Top row includes zero; bottom row conditions on positive exposure.

Narrower field definitions generate more markets and a larger share of zero-exposure programs, particularly under the Generic-field definition.

SUA exposure: event-study estimates

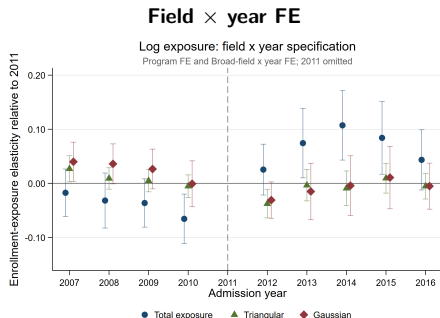
← First stage



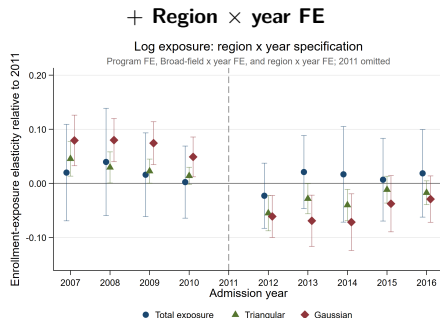
Notes: The figures use the Broad-field market definition and report separate event studies for Total, Triangular and Gaussian exposure. The left panel absorbs program and Broad-field × year fixed effects. The right panel additionally absorbs pre-treatment region × year fixed effects. Coefficients measure the enrollment difference associated with a 10 percentage-point increase in exposure in each year relative to 2011, the omitted year. Vertical bars report 95 % confidence intervals and standard errors are clustered by the pre-treatment market. Joint pretrend *p*-values for Total, Triangular and Gaussian exposure are 0.024, 0.568 and 0.020 in the left panel, and 0.285, 0.385 and 0.368 in the right panel.

SUA exposure: log event-study estimates

← Log first stage



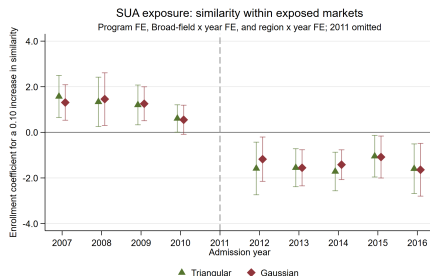
Zero exposure retained; D x year controls included and not plotted.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Total = 0.034; Triangular = 0.101; Gaussian = 0.100.



Zero exposure retained; D x year controls included and not plotted.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Total = 0.857; Triangular = 0.067; Gaussian = 0.000.

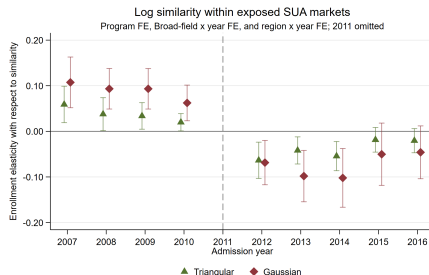
Notes: The figures use the Broad-field definition and report separate event studies for Total, Triangular, and Gaussian exposure. The left panel absorbs program and Broad-field × year fixed effects; the right panel additionally absorbs pre-treatment region × year fixed effects. For each exposure measure, zero-exposure programs are retained by setting $\log \widetilde{E_p^k} = 0$ when $E_p^k = 0$ and including $\mathbf{1}\{E_p^k > 0\} \times \mathbf{1}\{t = s\}$ separately for every event year. The plotted coefficients correspond to $\log \widetilde{E_p^k} \times \mathbf{1}\{t = s\}$ and are interpreted relative to 2011, the omitted year. Observations with zero first-year enrollment are excluded because the dependent variable is in logs. Vertical bars report 95 % confidence intervals; standard errors are clustered by the pre-treatment market. Joint tests of the 2007–2010 log-exposure coefficients yield pretrend p -values of 0.034, 0.101, and 0.100 for Total, Triangular, and Gaussian exposure in the left panel, and 0.857, 0.067, and 0.0001, respectively, in the right panel.

Levels



Exposed markets; economically meaningful zero similarity retained.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Triangular = 0.0209, Gaussian = 0.0078.

Logs

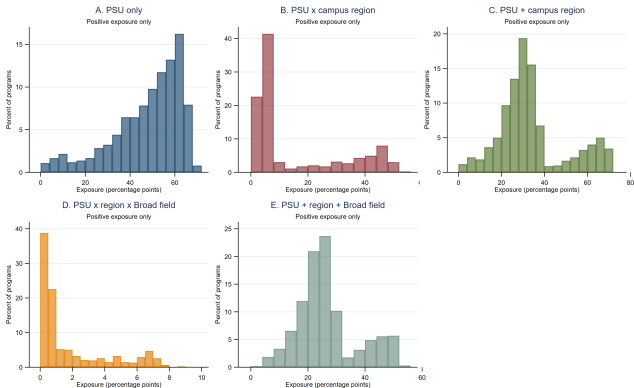


Exposed markets; Q = 0 retained; D x year controls included and not plotted.
95% confidence intervals; standard errors clustered by pre-treatment market.
Joint pretrend p-values: Triangular = 0.0348, Gaussian = 0.0001.

Notes: Coefficients are relative to 2011. All models absorb program, Broad-field $\times$ year, and pre-treatment region $\times$ year fixed effects; standard errors are clustered by pre-treatment market. The sample is restricted to markets with positive entrant presence, $M_m > 0$. Levels retain economically meaningful observations with $Q_p^k = 0$. In logs, zero-similarity observations are

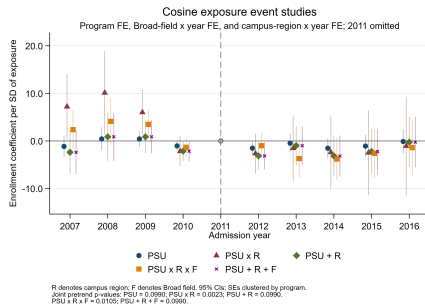
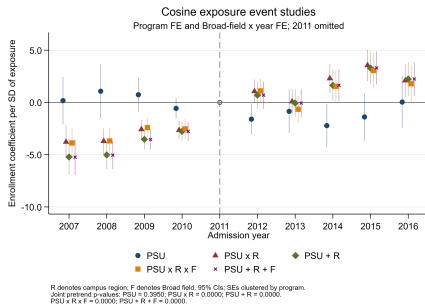
also retained by setting $\widetilde{\log Q_p^k} = 0$ when $Q_p^k = 0$ and including $\mathbf{1}\{Q_p^k > 0\} \times \mathbf{1}\{t = s\}$ separately in each event year; these indicator coefficients are not plotted. Observations with zero first-year enrollment are excluded from the log specification. Joint pretrend p-values are 0.0209 and 0.0078 for the triangular and Gaussian measures in levels, and 0.0348 and 0.0001 in logs. All four tests reject the absence of differential pre-treatment trends at the 5 % level; estimates are therefore interpreted as descriptive associations rather than causal effects.

Cosine exposure distributions One observation per incumbent program



Positive exposure only. Exposure measured in percentage points; each panel uses its own x-axis scale.

- Programs with zero exposure are excluded, allowing the shape of the positive part of each distribution to be seen more clearly.
- The multiplicative $\text{PSU} \times \text{campus region} \times \text{Broad field}$ measure remains concentrated at relatively low exposure values even after conditioning on positive exposure.



- Without campus-region $\times$ year FE, PSU only does not reject joint pretrends ($p = 0,395$), but all four measures that incorporate campus region do ($p < 0,001$).
- Adding campus-region $\times$ year FE improves the diagnostic for PSU and the additive measures (all $p = 0,099$), but the multiplicative measures still reject: PSU $\times$ region ($p = 0,0023$) and PSU $\times$ region $\times$ field ($p = 0,0105$).
- Post-2012 estimates become mostly negative and imprecise once campus-region $\times$ year FE are included. The min-10 sample is not shown and delivers virtually the same conclusions.
- Overall, the cosine results are best interpreted as descriptive rather than causal.